

4. Calculus

Ref	Content	Notes
Differentiation		
4.1	Know that the gradient function $\frac{dy}{dx}$ gives the gradient of the curve and measures the rate of change of y with respect to x	
4.2	Know that the gradient of a function is the gradient of the tangent at that point.	
4.3	Differentiation of kx^n where n is an integer, and the sum of such functions	Including expressions which need to be simplified first Given $y = (3x + 2)(x - 3)$ work out $\frac{dy}{dx}$ Given $y = \frac{5}{x^3}$ work out $\frac{dy}{dx}$
4.4	The equation of a tangent and normal at any point on a curve	
4.5	Increasing and decreasing functions	When the gradient is positive/negative a function is described as an increasing/decreasing function
4.6	Understand and use the notation $\frac{d^2y}{dx^2}$	Know that $\frac{d^2y}{dx^2}$ measures the rate of change of the gradient function
4.7	Use of differentiation to find maxima and minima points on a curve	Determine the nature either by using increasing and decreasing functions or $\frac{d^2y}{dx^2}$
4.8	Using calculus to find maxima and minima in simple problems	$V = 49x + \frac{81}{x} \quad x > 0$ Use calculus to show that V has a minimum value and work out the minimum value of V
4.9	Sketch/ interpret a curve with known maximum and minimum points	